C201/C255/I211/A506 Laboratory Session #8

Lab Goals – 1) Practice in using recursion in Java.

Helpful Handouts and programs:

1) Class Notes on recursion
2) Class program examples dealing with recursion. (found in programs section)
   a. Recursive1.java
   b. Recursive2.java

Create new program called *Lab8.java*. The purpose of this program is to write a recursive function that will find the sum of the elements of an integer array. In the main, create an integer array and initialize with {2,4,6,8,0}. Then call a recursive function that will return the sum of elements of the array.

Output shown here:

```
Sum of array elements is 20
```

Then change array to {2,4,6,8,7} and rerun and should get:

```
Sum of array elements is 27
```

Remember to do the two parts to avoid infinite recursion:

1) Recursive call to itself, with smaller version of same task.
2) One case with no recursive call, this is the stopping case.

Remember to also break the task down into two subtask.

1) Simple case.
2) Recursive case.

There should be no global variables used or needed to do this lab. The recursive function will need two parameters.

Write out an algorithm and hand trace through your logic to make sure it is working. You can always start with smaller array for hand tracing.

Make sure your name is in the top comment section. When done your source code should have at the bottom, in a comment section, the output from the program. When done let me know and I will take a look at it. For online class need to email the PDF to me.